Practical No 03

A) Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="409dp"
        android:layout_height="729dp"
        android:orientation="vertical"
        tools:layout_editor_absoluteX="1dp"
        tools:layout_editor_absoluteY="1dp">

        <TextView
            android:id="@+id/textView3"
            android:layout_width="match_parent"
            android:layout_height="38dp"
            android:text="Student Registration"
            android:textAlignment="center"
            android:textColor="#B10909"
            android:textColorHighlight="#5F0606"
            android:textColorLink="#731010"
            android:textIsSelectable="true"
            android:textSize="34sp" />

        <EditText
            android:id="@+id/editTextText6"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:ems="10"
            android:inputType="text"
            android:text="Enter Name"
            android:textColor="#062392" />
```

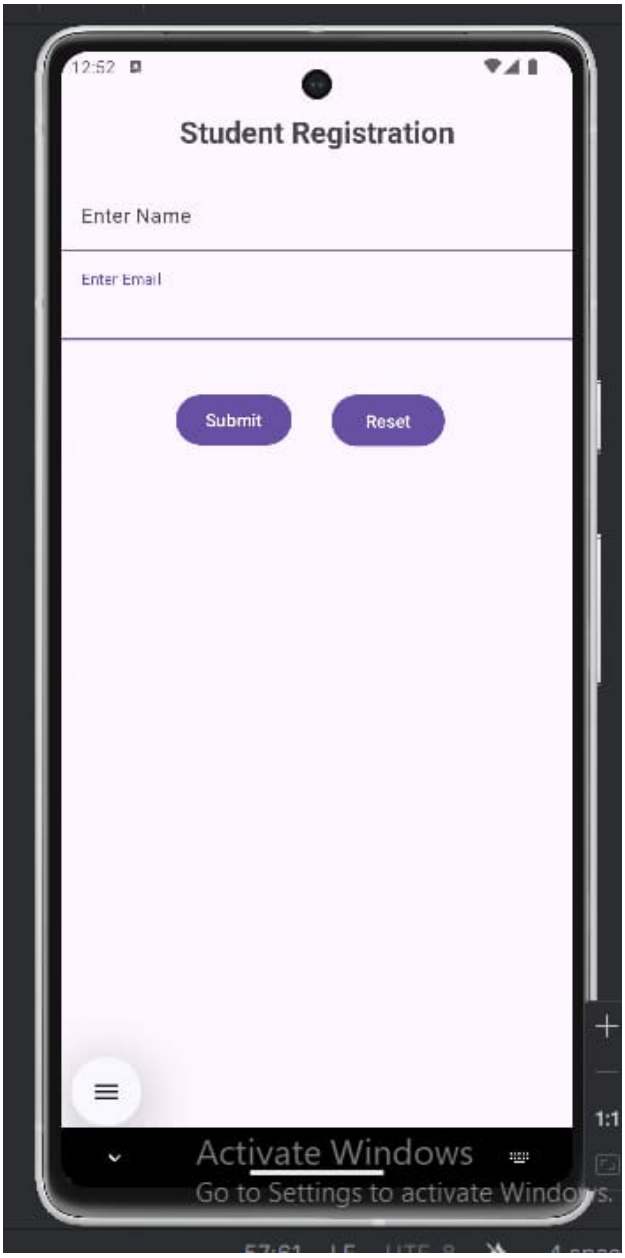
```
        <EditText
            android:id="@+id/editTextText7"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:ems="10"
            android:inputType="text"
            android:text="Enter Email"
            android:textColor="#00053A" />

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:orientation="horizontal">

            <Button
                android:id="@+id/button6"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:text="Submit" />

            <Button
                android:id="@+id/button7"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:text="Reset" />
        </LinearLayout>
    </LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:


12:52
Student Registration
Enter Name
Enter Email
Submit
Reset
Activate Windows
Go to Settings to activate Windows.

B) Develop an Android application using RelativeLayout to align UI components dynamically.

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layoutDirection="inherit"
    tools:context=".MainActivity">

    <RelativeLayout
        android:layout_width="409dp"
        android:layout_height="729dp"
        tools:layout_editor_absoluteX="1dp"
        tools:layout_editor_absoluteY="1dp">

        <TextView
            android:id="@+id/textView4"
            android:layout_width="372dp"
            android:layout_height="wrap_content"
            android:layout_centerHorizontal="true"
            android:text="RelativeLayoutDemo"
            android:textAlignment="center"
            android:textColor="#3E0000"
            android:textSize="34sp" />

        <ImageView
            android:id="@+id/imageView2"
            android:layout_width="165dp"
            android:layout_height="140dp"
            android:layout_centerHorizontal="true"
            android:layout_marginVertical="65dp"
            app:srcCompat="@android:drawable/btn_star_big_on"
            tools:srcCompat="@tools:sample/avatars" />

        <EditText
            android:id="@+id/editTextText4"
```

```
        android:layout_width="400dp"
        android:layout_height="wrap_content"
        android:layout_marginVertical="200dp"
        android:ems="10"
        android:inputType="text"
        android:paddingVertical="20dp"
        android:text="Enter Name"
        android:textColor="#040000" />

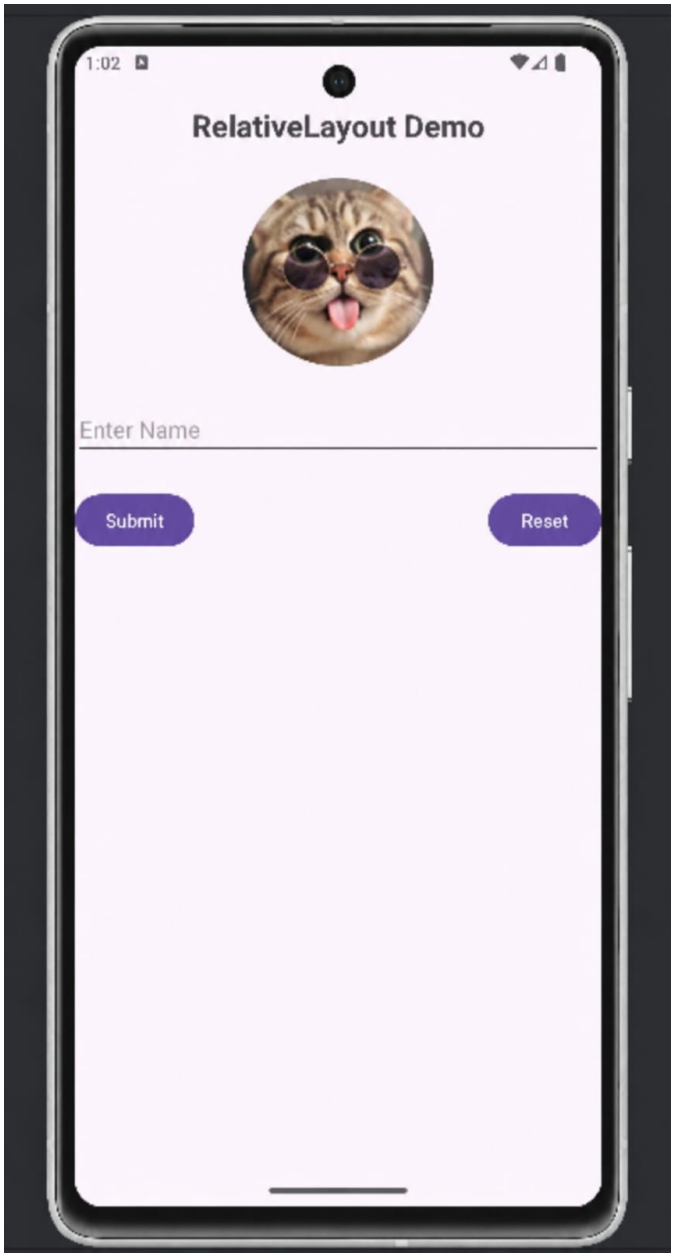
    <Button
        android:id="@+id/button4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginVertical="300dp"
        android:text="Button" />

    <Button
        android:id="@+id/button5"
        android:layout_width="100dp"
        android:layout_height="wrap_content"
        android:layout_marginVertical="300dp"
        android:text="SUBMIT" />

    <Button
        android:id="@+id/button6"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="false"
        android:layout_marginHorizontal="300dp"
        android:layout_marginVertical="300dp"
        android:text="RESET" />

  </RelativeLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



C) Develop an Android application using TableLayout to design a structured Login/Registration Form.

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    android:id="@+id/main"
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TableLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:stretchColumns="1">

        <!-- Heading -->
        <TableRow>

            <TextView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_span="2"
                android:gravity="center"
                android:text="Student Registration"
                android:textSize="28sp"
                android:textStyle="bold"
                android:padding="16dp"/>
        </TableRow>

        <!-- Name -->
        <TableRow>
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name"
        android:textSize="18sp"
        android:padding="8dp"/>

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"
        android:inputType="textPersonName"/>
</TableRow>

<!-- Email -->
<TableRow>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Email"
        android:textSize="18sp"
        android:padding="8dp"/>

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"/>
</TableRow>

<!-- Password -->
<TableRow>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Password"
        android:textSize="18sp"
        android:padding="8dp"/>
```

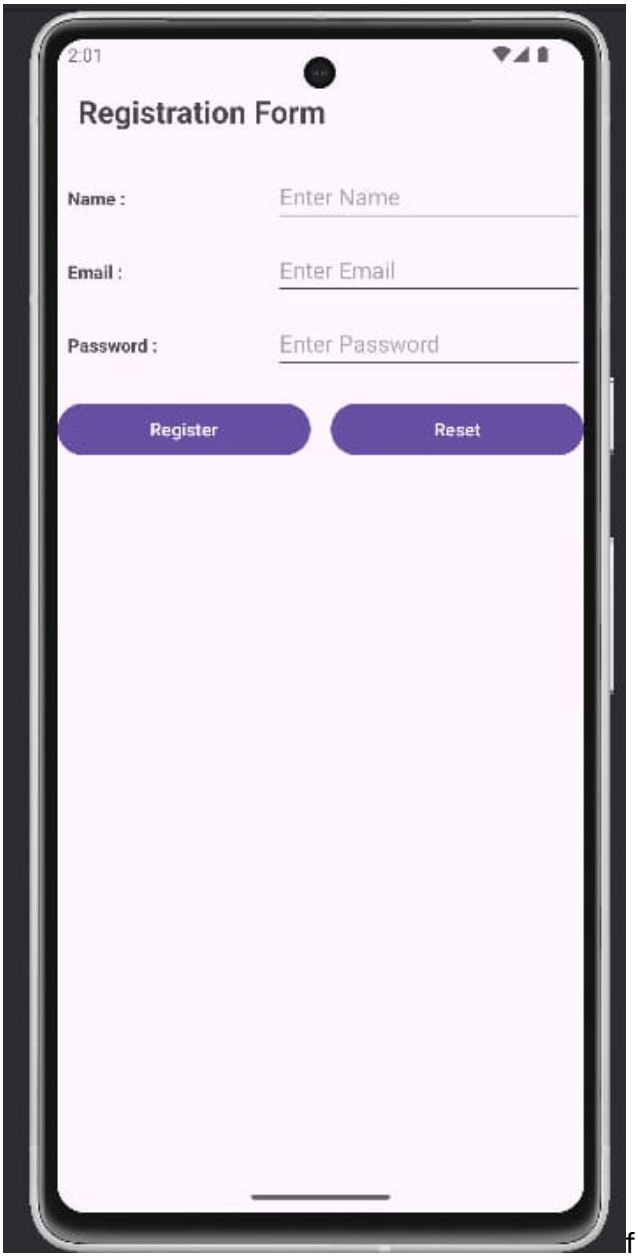
```
        <EditText
            android:id="@+id/etPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Password"
            android:inputType="textPassword"/>
    </TableRow>

    <!-- Buttons -->
    <TableRow
        android:gravity="center"
        android:paddingTop="20dp">

        <Button
            android:id="@+id/btnRegister"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Register"/>

        <Button
            android:id="@+id/btnReset"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="20dp"
            android:text="Reset"/>
    </TableRow>
  </TableLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:


2:01
Registration Form
Name :
Enter Name
Email :
Enter Email
Password :
Enter Password
Register
Reset